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entirely on data.

Fuzzy logic has emerged as a mathematical tool to deal with the uncertainties in human perception and reasoning. It provides a framework for an inference mechanism that allows for approximate human reasoning capabilities to be applied to knowledge-based systems. On the other hand, artificial neural networks have emerged as fast computation tools with learning and adaptive capabilities. These two fields have been integrated into a new emerging technology called fuzzy neural networks or neuro fuzzy system that combines the benefits of each field.

Q2. What is the difference between online and offline UPS?

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A2. *Offline UPS.* The offline/standby UPS offers battery backup. With this type of UPS, the load is normally connected directly to the

onds. Block diagrams of an offline UPS running on power supply and on power failure are shown in Fig. 1.

Online UPS. Online UPS, sometimes also called 'double-conversion UPS,' under normal operation always run off the battery, using their inverter for supply of AC power to the load. Therefore in this configuration, there are no power transfer switches and no transfer time in the event of a power failure. Thus this is truly an uninterrupted system.

To maintain the charge of the battery, a battery charging unit is incorporated in the system, which is continuously powered from the AC mains. This design means that if the mains power fails, power to the load is supplied from the inverter that is connected to the battery. Charging of the battery, however, will stop. This is what happens when a plugged-in laptop keeps on running without interruption when mains power fails.

The main advantage of the online UPS is their ability to provide an electrical fire-wall between the incoming power and sensitive electronic equipment.

While the standby and line-interactive UPS merely filter the input utility power, the double-conversion UPS provide a layer of insulation from power quality problems.

It allows control of the output voltage and frequency

regardless of the input voltage and frequency. Block diagram of an online UPS is shown in Fig. 2.

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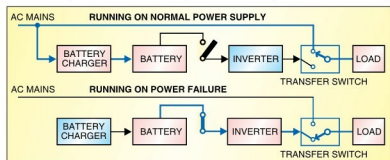


Fig. 1: Block diagram of an offline UPS

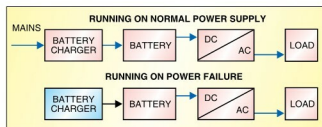


Fig. 2: Block diagram of an online UPS

incoming AC mains. When the incoming AC mains fails or falls below a predetermined level, the offline UPS turns on its internal DC-AC inverter circuitry, which is powered from an internal storage battery. The UPS mechanically switches the connected equipment onto its DC-AC inverter output. The switch-over time can be as long as 25 milliseconds.